Range

1 Find the range of the following sets of scores:

a $10, 7, 2, 14, 13, 15, 11, 4$

b $15, -2, -8, 8, 15, 6, -16, 15$

c $-0.5537, 1.7444, -0.3381, 0.7200, -0.3381, 1.0435$

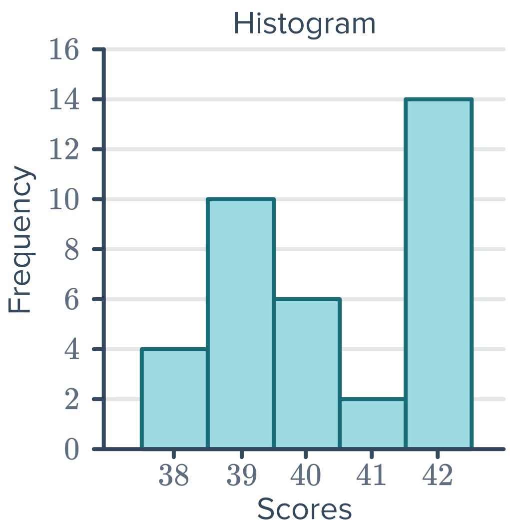
2 Consider the data provided in the table:

a Find the range of the scores.

b Find the mode.

Score	Frequency
68	16
69	41
70	30
71	31
72	49
73	29

3 Calculate the range for the data in the following histogram:



4 A group of students had a range in scores of 14 and the least score was 9? What was the greatest score in the group?

5 The range of a set of scores is 8, and the greatest score is 19. What is the least score in the set?

6 In a study, a group of people were shown [redacted]mes, and after 1 minute they were asked to recite as many names by memory as possible. The results are presented in the dot plot:

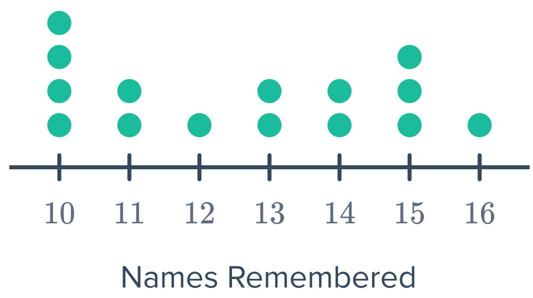
a) What does each dot represent?

b) How many people took part in the study?

c) What is the largest number of names someone remembered?

d) What was the smallest number of names someone remembered?

e) What is the range?



Interquartile range

7 For each set of scores, find:

i) The total number of scores

ii) The median

iii) The range

iv) The first quartile

v) The third quartile

vi) The interquartile range

a) $13, 15, 5, 16, 7, 20, 12$

b) $-3, -3, 1, 9, 9, 6, -9$

c) $33, 38, 50, 12, 33, 48, 41$

d) $1.2, 2.9, 3.5, 1.6, 3.2, 4.8, 1.9$

e)

	Leaf
2	2 5 6 7 9
3	0 0 5 6 8
4	0 0 1 8 9

Key: $5|2 = 52$

f)

Score	Frequency
5	3
13	3
16	2
28	2
31	3
38	4
48	2

8 For each data set represented by a dot plot below, find:

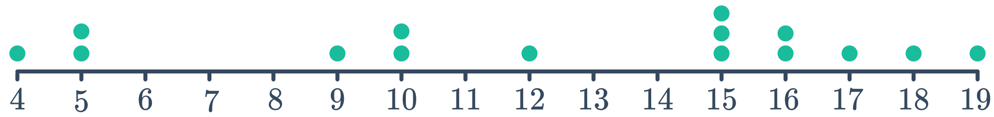
i The median

ii The first quartile

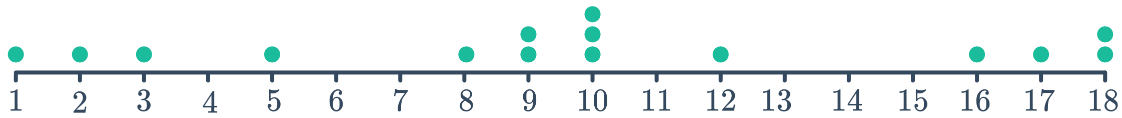
iii The third quartile

iv The interquartile range

a

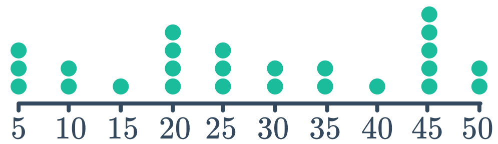


b



c

Sit-ups



d

Age of customers

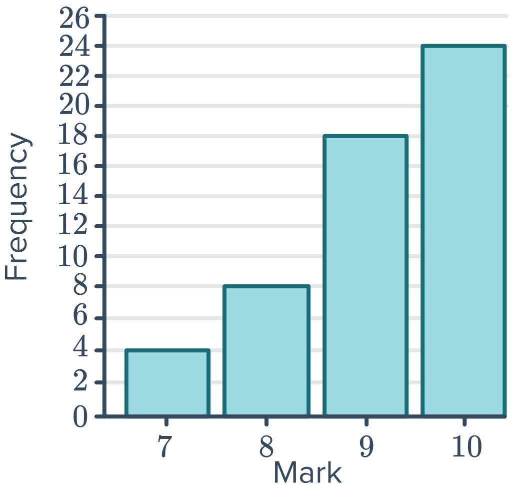
9 For each set of data, find the following:

i The first quartile

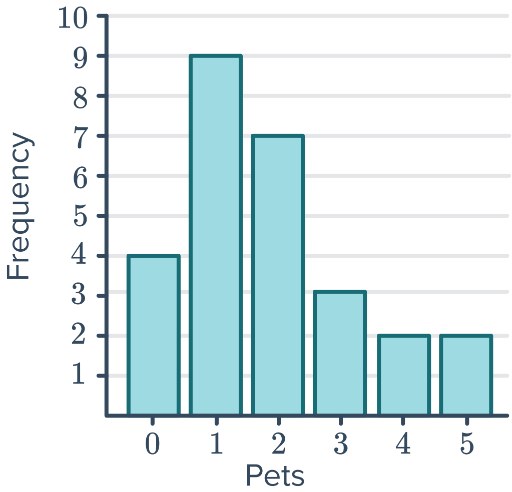
ii The third quartile

iii The interquartile range

a



b



c

	Leaf
6	2 7
7	1 2 2 4 7 9
8	0 1 2 5 7
9	0 1

Key: $5|2 = 52$

10 The data below shows the results of a survey conducted on the price of concert tickets locally and the price of the same concerts at an international venue:

a Find the interquartile range for the international venue.

b Find the interquartile range for the local venue.

c At which venue is there the least spread in the middle 50% of prices?

International		Local
9 9 9	**6**	8
8 5 5 5 3 0	**7**	5 6 6 9 9
8 4 3 2 1 0	**8**	2 2 6 6
5 3 2 0	**9**	0 0 1 4 5 6 8
5	**10**	0 3 5

Key: $9|6|8 = \$69$ and $\$68$

11 A group of students were asked how many phone calls they had made the previous day. The information was collected in the given frequency table:

Phone calls	Frequency
0	8
1	5
2	10
3	6
4	6
5	7
6	3
7	2

a How many students were surveyed?

b Find the range of the data.

c Find the interquartile range.

12 A sample of boxes of matches were selected for quality control and the number of matches in each box recorded in the given frequency table:

Matches	Frequency
45	1
46	2
47	5
48	3
49	7
50	25
51	6
52	2

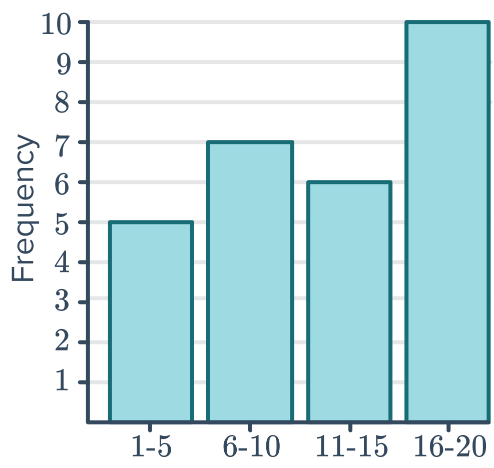
a How many matchboxes were sampled?

b Find the range of the data.

c Find the interquartile range.

13 Consider the given column graph:

Class	Class center	Frequency
$1-5$		
$6-10$		
$11-15$		
$16-20$		



a Complete the given frequency table.

b Use the class centers to estimate the:

i Range

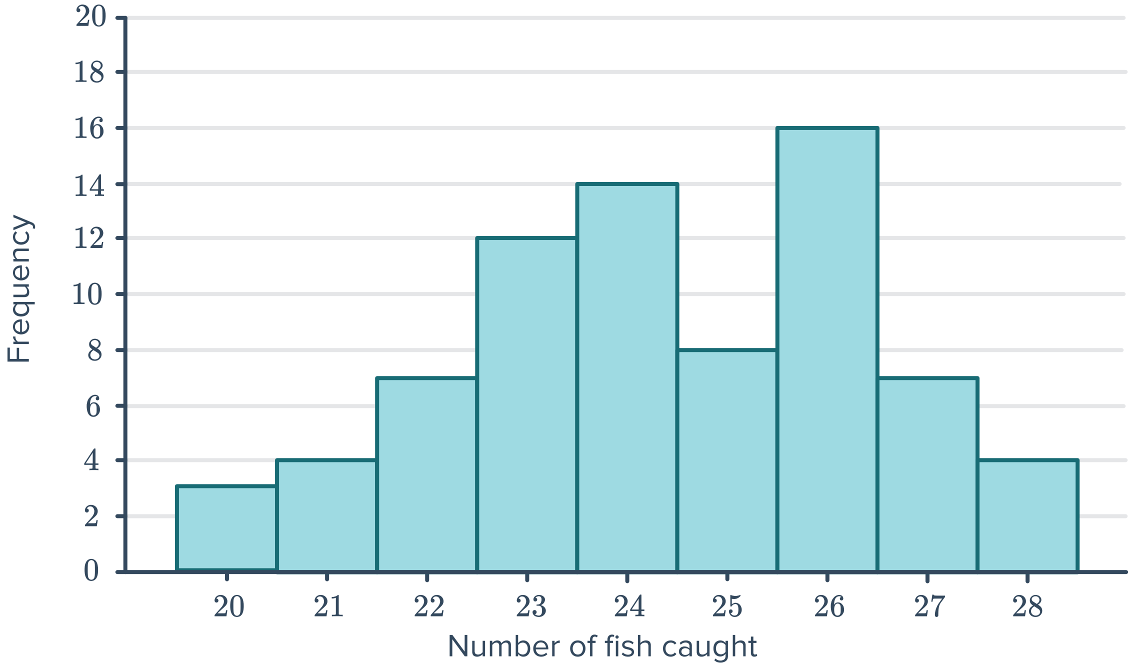
ii Interquartile range

14 Edward records the number of fish he cat[redacted]or each fishing trip over a period of time:

a How many fishing trips did he go on?

b Find the range in the number of fish caught.

c Find the interquartile range.



15 In competition, a diver must complete 8 rounds of dives. Her scores for the first 7 rounds are:

$$7.3,\ 7.4,\ 7.7,\ 8.4,\ 8.7,\ 8.9,\ 9.4$$

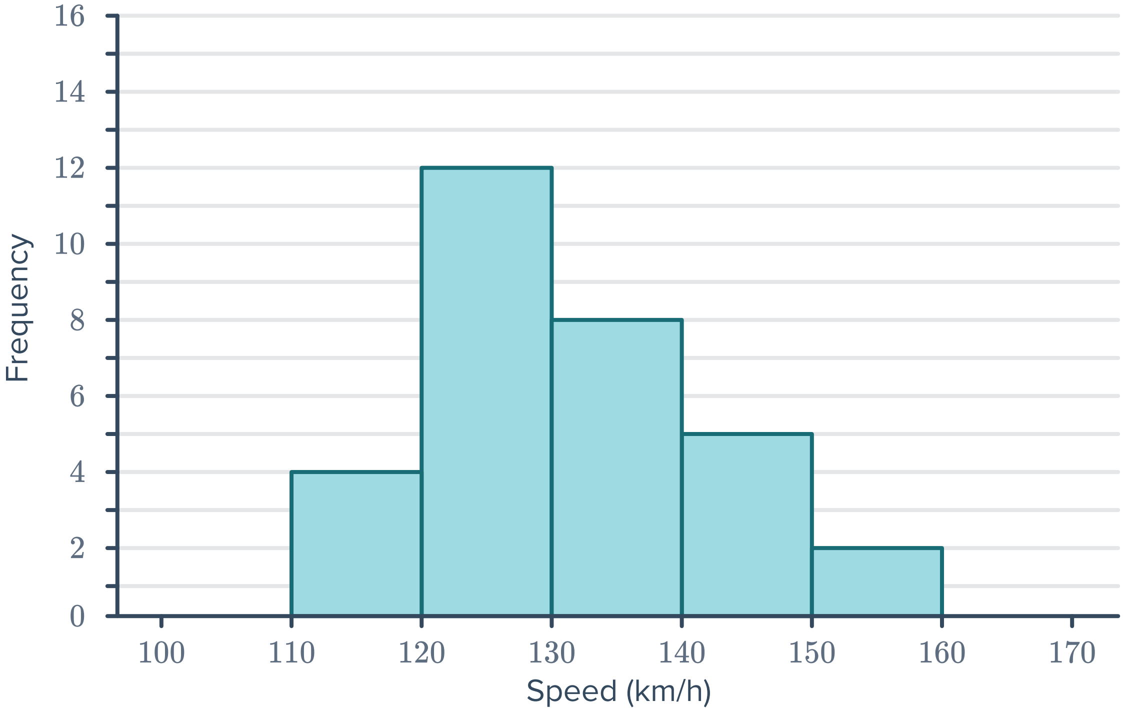
Find her score in the 8th round if the upper quartile of all of her 8 scores is 8.85.

16 In a competition, a contestant must complete 12 challenges earning as many points as possible. Her scores for the first 11 challenges are:

$$32,\ 38,\ 45,\ 55,\ 66,\ 86,\ 94,\ 98,\ 106,\ 108,\ 117$$

Find her score in the 12th round if the lower quartile of all of her 12 scores is 47.5.

17 The histogram below displays the speeds [redacted]ls delivered by a bowler in cricket:



a Create a frequency table for the data and include a column for class centers.

b Use the class centers to estimate the:

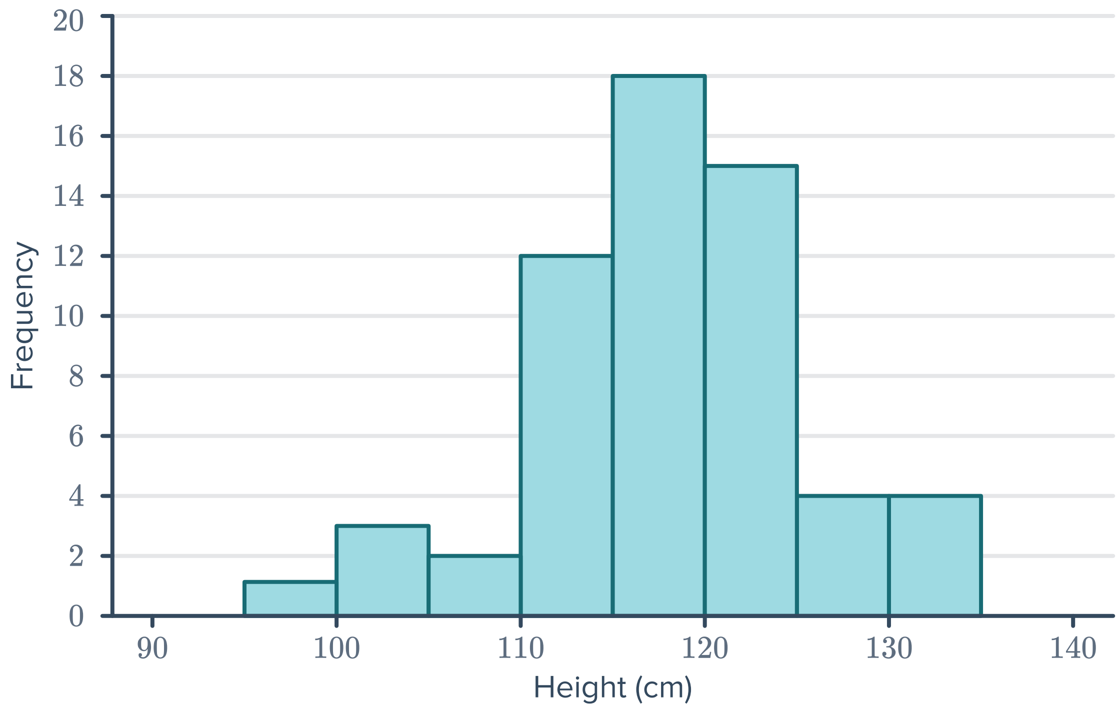
i Range

ii First quartile

iii Third quartile

iv Interquartile range

18 The following histogram shows the heights of tomato plants in a greenhouse.



b   Use the class centers to estimate:

i   The range

ii   The interquartile range

19 In a survey the mass (in grams) of 30 individual apples from an orchard were noted and recorded in the following list:

$$86,\quad 87,\quad 91,\quad 93,\quad 94,\quad 95,\quad 96,\quad 96,\quad 98,\quad 99$$

$$100,\quad 101,\quad 102,\quad 103,\quad 103,\quad 104,\quad 104,\quad 105,\quad 106,\quad 106$$

$$106,\quad 107,\quad 107,\quad 107,\quad 108,\quad 108,\quad 109,\quad 109,\quad 109,\quad 109$$

a   Find the following statistics:

i   The smallest value

ii   The largest value

iii   The range

iv   The first quartile

v   The third quartile

vi   The interquartile range

b   Complete the given frequency table.

c   Using the class centers, find an estimate for:

i   The range

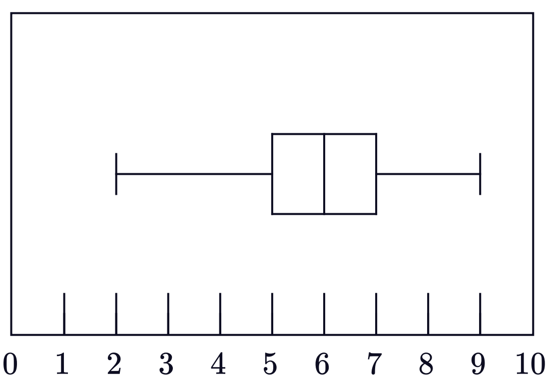
ii   The interquartile range

d   How much less was the approximation for the interquartile range than the actual interquartile range?

Weight (g)	Class center	Frequency
$85 \leq x < 90$	87.5	
$90 \leq x < 95$		3
$95 \leq x < 100$		
$100 \leq x < 105$		
$105 \leq x < 110$		

Box plots

20 What is the range of the data represented by the given box plot?

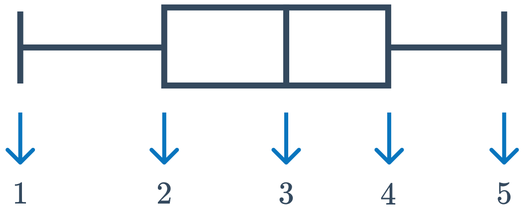


21 Construct a box plot for the five number summary:

- Median $= 35$
- Lower Quartile $= 25$
- Upper Quartile $= 60$
- least score $= 5$
- greatest score $= 75$

22 Consider the box plot and statistics listed below:

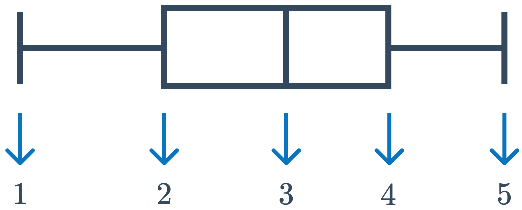
- Median $= 47$
- Lower Quartile $= 33$
- Upper Quartile $= 61$
- least score $= 16$
- greatest score $= 71$



Which number should go in position 2 on the box plot?

23 Consider the box plot and statistics listed below:

- Median $= 36$
- Lower Quartile $= 28$
- Upper Quartile $= 42$
- least score $= 20$
- greatest score $= 52$



Which number should go in position 4 on the box plot?

24 Using the information in the table, create a box plot to represent this data set:

Minimum	5
Lower Quartile	25
Median	35
Upper Quartile	60
Maximum	75

25 A geography teacher has marked a set of tests. She wants to represent the results in a box plot. She has already sorted her data and created the table shown:
Create a box plot to match the data in the table.

Minimum	8
Lower Quartile	10
Median	16
Upper Quartile	24

26 Consider the following data set:

$$20,\ 36,\ 52,\ 56,\ 24,\ 16,\ 40,\ 4,\ 28$$

a Find the five number summary.

b Construct a box plot for the data.

27 The table shows the luggage weight, in kilograms, of 30 passengers.

a What is the mean check in weight? Round your answer to two decimal places.

b Determine the median, Q_1, and Q_3.

c In which quartile does the mean lie?

Weight	Frequency
16	5
17	5
18	2
19	4
20	6
21	4
22	4

28 For the box plot shown, find the following:

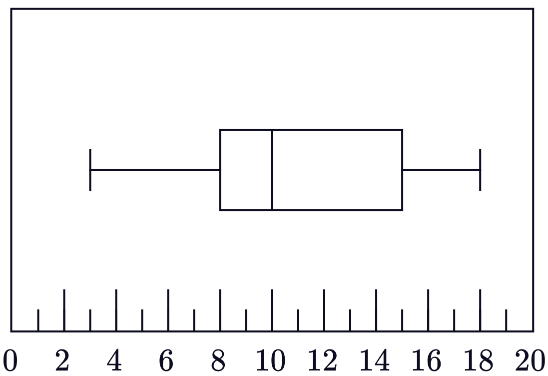
a Minimum score

b Maximum score

c Range

d Median

e Interquartile range



29 Consider the box plot shown:

a State the percentage of scores that lie between each of the following values:

i 7 and 15

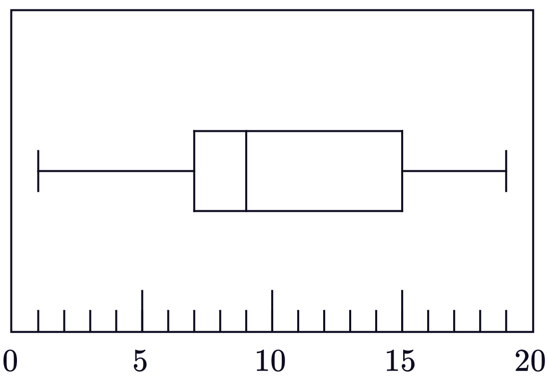
ii 1 and 7

iii 19 and 9

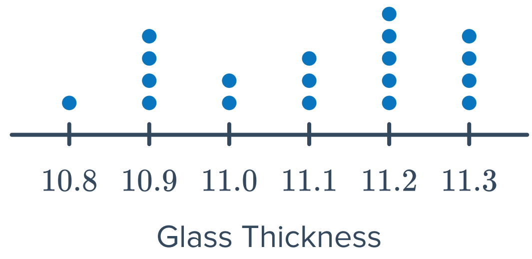
iv 7 and 19

v 1 and 15

b In which quartile is the data the least spread out?



30 The glass windows for an airplane are cut [illegible]ertain thickness, but machine production means there is some variation. The thickness of each pane of glass produced is measured (in millimeters), and the dot plot shows the results:



a Find the median thickness, to two decimal places.

b Find the interquartile range.

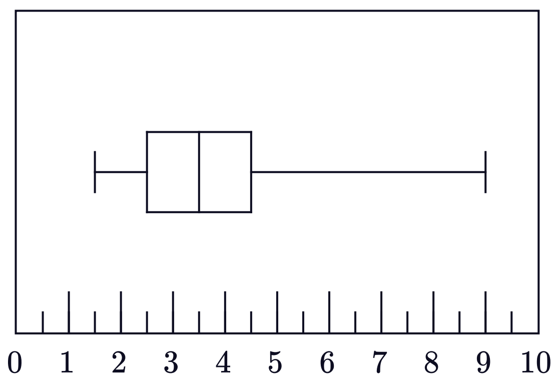
c Construct a box plot to represent the data.

d What percentage of thicknesses were between 10.8 mm and 11.2 mm inclusive? Round your answer to two decimal places.

e According to the box plot, in which quartile are the results the most spread out?

f Which statistics cannot be found from a box plot?

31 In training, a fighter pilot measures the number of seconds he blacks out over a number of flights. He constructs the following box and whisker plot for his data: As long as the pilot is not unconscious for more than 7 seconds, he will be safe to fly. The pilot concludes that he is safe to fly all the time. Is his conclusion correct? Explain your answer.



32 The box plot shows the age at which a group of people got their driving licenses:

a What is the oldest age?

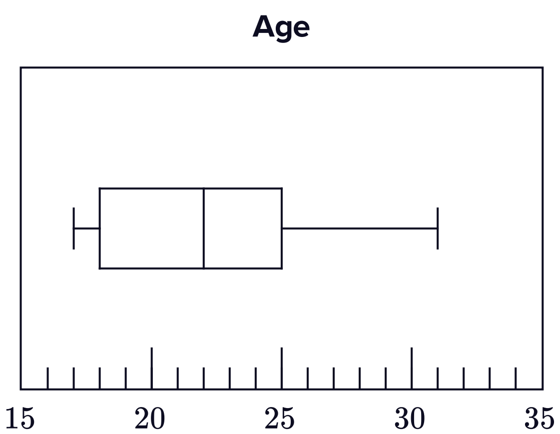
b What is the youngest age?

c What percentage of people were aged from 18 to 22?

d The middle 50% of responders were within how many years of one another?

e In which quartile are the ages least spread out?

f The bottom 50% of responders were within how many years of one another?



Standard deviation

33 For each of the given sets of data find the following correct to two decimal places:

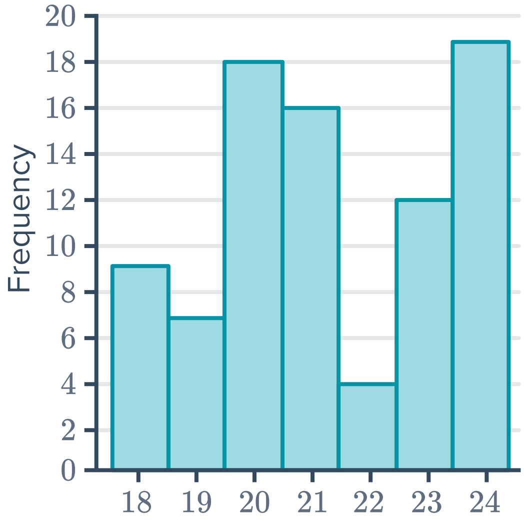
i Population standard deviation

ii Sample standard deviation

a $8, 20, 16, 9, 9, 15, 5, 17, 19, 6$

b $-17, 2, -6, 9, -17, -9, 3, 8, 5$

c



d

	Leaf
1	0 2 6
2	6 8
3	0
4	5 5 6
5	6 6
6	0
7	7

Key: $1|2 = 12$

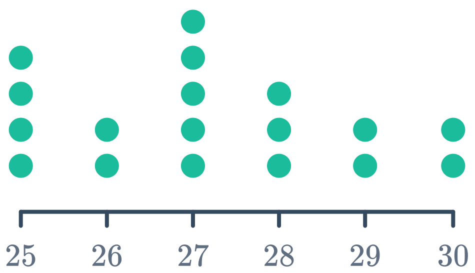
e

Score	Frequency
15	13
16	9
17	23
18	19
19	8
20	13

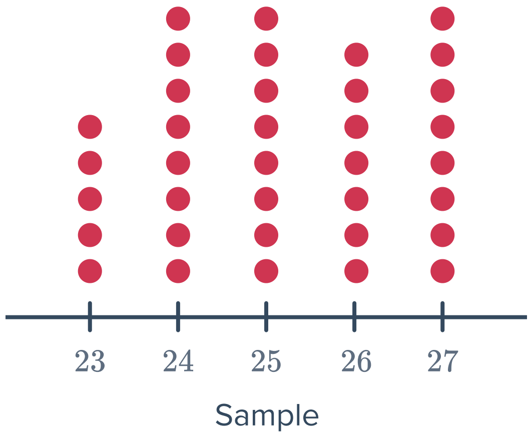
f

Score	Frequency
68	16
69	41
70	30
71	31
72	49
73	29

g



h



34 The scores of five diving attempts by a professional diver are recorded below:

$$6.4,\ 5.9,\ 5.1,\ 5.1,\ 5.5$$

a Calculate the sample standard deviation of his scores. Round your answer to two decimal places.

b On his sixth attempt, the diver scores 9.3. State whether this score would increase or decrease the following:

i The mean   **ii** The standard deviation

c If each judge gave the diver the same score, what would be the sample standard deviation of the judges' scores?

35 The trial times (in seconds) of an athlete for the $100\ \text{m}$ sprint are recorded below:

$$11.0,\ 10.5,\ 12.2,\ 11.2,\ 11.6,\ 11.7,\ 10.8,\ 12.1,\ 11.0,\ 10.9$$

a Calculate the sample standard deviation of her times. Round your answer to two decimal places.

b On her next attempt, she manages to run $100\ \text{m}$ in $10.1\ \text{s}$. Explain what impact this latest attempt would have on her mean and standard deviation.

36 A batsman's mean number of runs is 62 and the standard deviation is 13. In the next match he makes 50 runs. Explain what impact this latest match would have on his mean and standard deviation.

37 Meteorologists predicted huge variation in temperatures throughout the month of April. The temperature each day for the first two weeks of April was recorded in degrees celsius:

$$16,\ 16.5,\ 16.5,\ 17.5,\ 19.5,\ 22.5,\ 23.5,\ 23.5,\ 26,\ 26,\ 26.5,\ 27,\ 27.5,\ 28$$

a What is the range of the temperatures?

b What is the interquartile range of the temperatures?

c What is the sample standard deviation? Round your answer to one decimal place.

d Would the sample standard deviation or the interquartile range be the best measure of spread to support or counter a prediction? Explain your answer.

38 In an entrance exam, applicants completed two papers. On average, students performed better in Paper 1, but their scores were less spread out in Paper 2.
Describe how the size of the mean and standard deviation of Paper 1 would compare to Paper 2.

39 Seven millionaires with an average net we[redacted]f $\$33$ million with a standard deviation of $\$7$ million are having a party. Then Carlos Slim, who has a net wealth estimated to be $\$43$ billion, enters the room.

a. What is the new average net wealth (in millions of dollars) in the room? Round your answer to the nearest million dollars.

b. When Carlos Slim's net worth is taken into account, will the standard deviation be greater, smaller or unchanged from before?

c. Will the range be greater, smaller or unchanged from before?

40 Consider the following set of scores:

$$19, 18, 14, 19, 10$$

a. Find the mean.

b. Find the range.

c. Complete the following table:

Score (x)	$x - \text{mean}$	$(x - \text{mean})^2$
19		
18		
14		
19		
10		

d. Find the sample standard deviation, correct to two decimal places.

41 Consider the following set of scores:

$$41, -15, -7, 10, 26$$

a Find the mean.

b Find the range.

c Complete the following table:

Score (x)	$x - \text{mean}$	$(x - \text{mean})^2$
-15		
-7		
10		
26		
41		

d Find the sample standard deviation, correct to two decimal places.